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A Generative AI System for Crop Disease Diagnosis and Localized Treatment Recommendation Using Multimodal LLMs and Farmer Voice Queries

A CAPSTONE PROJECT REPORT

A Capstone Project Report submitted in partial fulfilment of the requirements for the Course of

CSA6502 – Generative AI and Large Scale Model to
the award of the degree of

BACHELOR OF ENGINEERING & TECHNOLOGY Submitted by

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BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled “A Generative AI System for Crop Disease Diagnosis and Localized Treatment Recommendation Using Multimodal LLMs and Farmer Voice Queries” has been carried out by Surya Krishna Teja (192425065), Mounish A (192472273), and N. Somasekhar Naidu (192472347) under the supervision of Dr. Nithisha J and is submitted in partial fulfilment of the requirements for the current semester of the B.Tech & B.E program at program at Saveetha Institute of Medical and Technical Sciences, Chennai.

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DECLARATION

We, Surya Krishna Teja (192425065), Mounish A (192472273), and N. Somasekhar Naidu (192472347), hereby declare that the capstone project entitled “A Generative AI System for Crop Disease Diagnosis and Localized Treatment Recommendation Using Multimodal LLMs and Farmer Voice Queries” is the result of our academic project work carried out under the guidance of Dr. Nithisha J.

We further declare that the implementation, documentation, testing, and presentation of the project have been prepared for academic purposes and that the system is intended as a first-level support and classification tool rather than an official cybercrime investigation mechanism.

Place: SIMATS ENGINEERING, THANDALAM CAMPUS, CHENNAI

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Individual Contribution Statement

Student Name / Register No.	Specific Responsibilities	Design & Development Contribution	Testing & Analysis Contribution	Report Contribution	Approx. Contri. (%)
SURYA KRISHNA TEJA (192425065)	Module 2: Multimodal AI Diagnosis & Farmer Query Module.	Developed the multimodal diagnosis workflow, farmer text/voice query handling and integration of crop image context with the AI prompt.	Tested multimodal inputs, voice-query flow and diagnosis response behaviour.	Contributed to Chapters 3–5 and implementation evidence.	33%
MOUNISH A (192472273)	Module 1: Crop Information & Image Upload Module. Crop selection, location, season and imageupload foundation.	Designed and developed the Module 1 interface and workflow, including crop information entry, location/season selection, image upload and input validation.	Tested crop-information fields, image formats, required inputs and successful transition to AI diagnosis.	Contributed strongly to Chapter 1, Chapter 4 Module 1 implementation, screenshots, testing and appendix evidence.	34%
N. SOMASEKHAR NAIDU (192472347)	Modules 3–4: Disease Analytics & Recommendation and voice/output support.	Contributed to analytics presentation, disease distribution, recommendation flow and user-facing result presentation.	Tested dashboard values, recommendation display and integrated output behaviour.	Contributed to Chapters 4–6, results, conclusion and future enhancement.	33%
TOTAL	Comprehensive project execution across all functional modules.	Collaborative Python/Streamlit development, multimodal AI integration, interface design and module integration.	End-to-end functional and usability-oriented testing.	Complete technical report, implementation evidence and presentation prepared collaboratively.	100%

ABSTRACT

Agricultural productivity can be significantly affected when crop diseases are not identified early or when farmers cannot access suitable treatment guidance. The proposed Generative AI System for Crop Disease Diagnosis and Localized Treatment Recommendation combines crop information, plant images, farmer text or voice queries, Speech-to-Text processing, and multimodal large language models to provide an integrated assistance workflow.

The system accepts crop type, crop location, season, and a crop leaf, fruit, or plant image as the primary information for AI-based disease analysis. Farmer questions can be entered through text or voice. Speech-to-Text converts voice queries into text, while a multimodal AI model such as Gemini can interpret the image together with crop information and the farmer's question. The Generative AI layer then produces a simple, localized and context-aware response containing disease interpretation and treatment or preventive recommendations.

The project is organized into modules for crop information and image upload, multimodal AI diagnosis and farmer query, disease analytics and recommendation, and supporting implementation/result presentation. Module 1 is the primary individual contribution of Mounish A because it establishes the quality and completeness of the input context before AI analysis.

The design emphasizes usability, multilingual interaction, localized guidance and responsible AI use. Recommendations are intended as decision support and should be verified with suitable agricultural expertise, especially for pesticide or treatment decisions.

Keywords

Generative AI, Multimodal LLM, Crop Disease Diagnosis, Speech-to-Text, Farmer Voice Queries, Localized Treatment Recommendation, Gemini, Streamlit.

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CHAPTER 1: INTRODUCTION AND PROJECT OVERVIEW

1.1 Background

Crop diseases can reduce yield, quality and farmer income when symptoms are not recognized early. Farmers may have difficulty identifying diseases from visible symptoms and may not know which treatment or preventive action is appropriate for a specific crop, location or season.

Generative AI and multimodal large language models provide an opportunity to combine visual information with natural-language interaction. Instead of requiring the farmer to understand technical disease terminology, the proposed system accepts a crop image together with structured crop information and a text or voice query.

1.2 Project Title

A Generative AI System for Crop Disease Diagnosis and Localized Treatment Recommendation Using Multimodal LLMs and Farmer Voice Queries

1.3 Problem Statement

Existing crop disease applications commonly focus on image classification. Such systems may identify a disease but provide limited conversational support, localized treatment guidance or natural voice interaction. The project therefore addresses the problem of creating an integrated Generative AI application that can interpret crop images and farmer questions and generate understandable, context-aware assistance.

1.4 Aim

To design and implement a Generative AI-based crop disease assistance system that combines crop images, structured crop information and farmer voice/text queries to support disease diagnosis and localized treatment recommendation.

1.5 Objectives

- Allow farmers to select crop type, location and current season.
- Accept crop leaf, fruit or plant images in common formats.
- Accept farmer questions through text and voice input.
- Convert voice queries into text for AI processing.
- Use a multimodal LLM to interpret image and textual context together.
- Generate localized and crop-specific treatment and preventive recommendations.
- Present results in a simple and understandable interface.
- Provide analytics for diagnoses and disease distribution.
- Maintain a modular architecture for future expansion and testing.

1.6 Scope

- Crop information capture and image upload.
- Multimodal crop disease diagnosis support.
- Farmer voice and text interaction.
- Localized recommendation generation.
- Disease analytics and dashboard presentation.
- Future multilingual, weather, soil and mobile extensions.

1.7 System Modules

Module	Primary Responsibility	Priority in This Report
Module 1 – Crop Information & Image Upload	Crop type, location, season and image input; validation and preparation of context.	HIGH – Detailed

Module 2 – Multimodal AI Diagnosis & Farmer Query	Text/voice query, Speech-to-Text, multimodal prompt and diagnosis.	Supporting
Module 3 – Disease Analytics & Recommendation	Diagnosis statistics, confidence display, disease distribution and recommendation presentation.	Supporting
Module 4 – Voice / User Assistance	Voice-oriented interaction and simple userfacing output.	Supporting

Table 1.1: System Modules and Report Priority

Module 1 receives the highest level of documentation in this report because it was the primary individual contribution of Mounish A. The module creates the structured input context that is later consumed by the multimodal AI stage.

Module 1 – Detailed Contribution Focus

- Crop selection provides the AI with the plant category being examined.
- Location supports localized interpretation and future region-specific guidance.
- Season provides environmental context for diagnosis and recommendation.
- Image upload supplies the visual evidence for disease analysis.
- Input validation reduces incomplete or unsuitable requests before AI processing.

1.8 Responsible AI Considerations

- AI output should be treated as assistance rather than guaranteed agricultural diagnosis.
- Users should verify high-impact treatment decisions with qualified agricultural experts.
- The system should avoid presenting uncertain predictions as facts.
- Uploaded images and farmer information should be handled with appropriate privacy controls.

CHAPTER 2: PROBLEM IDENTIFICATION AND LITERATURE REVIEW

2.1 Problem Identification

The central problem is to support disease identification and treatment guidance from information that may be incomplete, informal or difficult for a farmer to describe technically. A multimodal system must combine visual evidence with crop type, location, season and the farmer's question.

2.2 Existing System

- Many existing solutions focus primarily on leaf-image classification.
- CNN and transfer-learning approaches are commonly used for disease recognition.
- Image classifiers generally provide a fixed disease label rather than a conversational explanation.
- Voice-based farmer interaction and regional-language support are limited.
- Localized treatment recommendation is often separated from disease detection.
- A single integrated workflow for image, voice, diagnosis and recommendation is less common.

2.3 Proposed System

- Structured crop information is collected before analysis.
- A crop image is uploaded as the main visual input.
- Voice queries are converted to text through Speech-to-Text.
- The image, crop information and query are combined for multimodal reasoning.
- Generative AI produces an understandable response with disease interpretation and recommendations.
- Analytics summarize diagnoses and disease distribution.

2.4 Literature Review Summary

Year	Study / Focus	Methods / Contribution	Gap Relevant to Project
2016	Mohanty et al. – Plant disease detection	Deep learning and PlantVillage image dataset	Image classification does not provide conversational localized treatment guidance.
2018	Ferentinos – Plant disease recognition	Deep learning / CNN classification	Focus is disease recognition rather than multimodal farmer assistance.
2019	Too et al. – Plant disease classification	Transfer learning with CNN architectures	Does not integrate voice queries and Generative AI responses.
2024	Generative AI / multimodal AI studies	Vision-language and multimodal models	Requires domain-specific integration with crop context and localized recommendations.

Table 2.1: Literature Review Summary

2.5 Engineering Gap

The engineering gap addressed by the project is the integration of multiple interaction and reasoning components into one farmer-oriented workflow. A disease image alone may not contain enough context for a useful recommendation. The proposed system therefore combines visual evidence with crop identity, geographic context, season and a natural-language question.

The supplied project presentation specifically positions the system as a multimodal LLM solution that combines disease diagnosis, voice interaction, localized recommendations and automated assistance. This integrated design is the main practical contribution of the project.

2.6 Need for Generative AI

- Natural-language generation makes results easier for non-technical users to understand.

- Multimodal models can reason over image and textual context in a single interaction.
- Generative responses can include preventive steps, treatment considerations and follow-up questions.
- Voice interaction can reduce the barrier of typing for field users.
- Localized context can be included in prompts to improve relevance.

2.7 Expected Engineering Benefits

Aspect	Expected Benefit
Accessibility	Voice and text interaction supports different user preferences.
Context	Crop, location, season and image are combined rather than treated independently.
Explainability	Generated responses can explain visible symptoms and recommended next steps in simple language.
Extensibility	Weather, soil, multilingual and mobile components can be integrated later.
Usability	Streamlit provides a simple interface for rapid deployment and demonstration.

Table 2.2: Engineering Benefits

CHAPTER 3: METHODOLOGY, DATA AND SYSTEM DESIGN

3.1 Development Methodology

The project follows a modular development approach so that input capture, multimodal reasoning, analytics and user assistance can be developed and tested independently. The overall workflow is: Crop Information → Image Upload → Farmer Query → Speech-to-Text (when voice is used) → Multimodal LLM Analysis → Disease Interpretation → Localized Recommendation → Result / Analytics. **3.2 Input Data and Context**

Input	Purpose	Example
Crop Type	Identifies the plant category	Tomato
Location	Provides regional context	Regional / district location
Season	Adds environmental context	Kharif / Rabi / current season
Crop Image	Provides visual disease evidence	Leaf / fruit / plant image
Farmer Query	Describes the concern or asks a question	“What is wrong with my leaf?”
Voice Query	Alternative to typing	Recorded farmer question

Table 3.1: Input Data and Context

3.3 Representative User Inputs

- Crop: selected by the farmer.
- Location: entered to support localized guidance.
- Season: selected from the available options.
- Image: uploaded in JPG, PNG or WEBP format.
- Question: entered as text or supplied through voice input.

3.4 Data Flow

Module 1 prepares the input context and performs the first validation stage. Only after the required information is available does the application proceed toward multimodal diagnosis. This separation makes Module 1 a critical foundation for reliable downstream AI processing.

3.4 Multimodal AI Processing Pipeline

Stage	Operation	Purpose
1	Input validation	Check crop information and image availability.
2	Voice capture	Accept farmer speech when voice interaction is selected.
3	Speech-to-Text	Convert spoken query into text.
4	Context assembly	Combine crop, location, season, image and question.
5	Multimodal LLM analysis	Interpret visual and textual evidence together.
6	Disease interpretation	Generate probable disease/symptom explanation.
7	Recommendation generation	Produce localized treatment and preventive guidance.
8	Response presentation	Display simple text and voice-oriented output where implemented.

Table 3.2: Multimodal AI Processing Pipeline

3.5 Prompt Construction and Response Generation

The multimodal request is designed to provide the model with structured context before asking for a diagnosis and recommendation. A typical logical prompt contains crop type, location, season, farmer question and the uploaded image. The response is expected to be concise, understandable and organized into disease interpretation, observed indicators, recommended actions and preventive measures.

- Use the supplied crop information as context.
- Ask the model to inspect the uploaded image for visible symptoms.
- Include the farmer's natural-language question.
- Request localized and season-aware recommendations when sufficient context is available.
- Avoid presenting uncertain results as guaranteed facts.

3.6 System Design Principles

- Modularity – each major function has a defined responsibility.
- Usability – interface elements are simple and direct.
- Context-awareness – structured fields supplement the image.
- Multimodality – image and natural language are processed together.
- Responsible generation – recommendations are framed as assistance.

3.6 System Architecture

The proposed architecture, reproduced from the project presentation, connects the farmer-facing input layer with multimodal AI diagnosis, recommendation and analytics. Module 1 supplies crop information and the image, while later modules process the query and produce the final assistance.

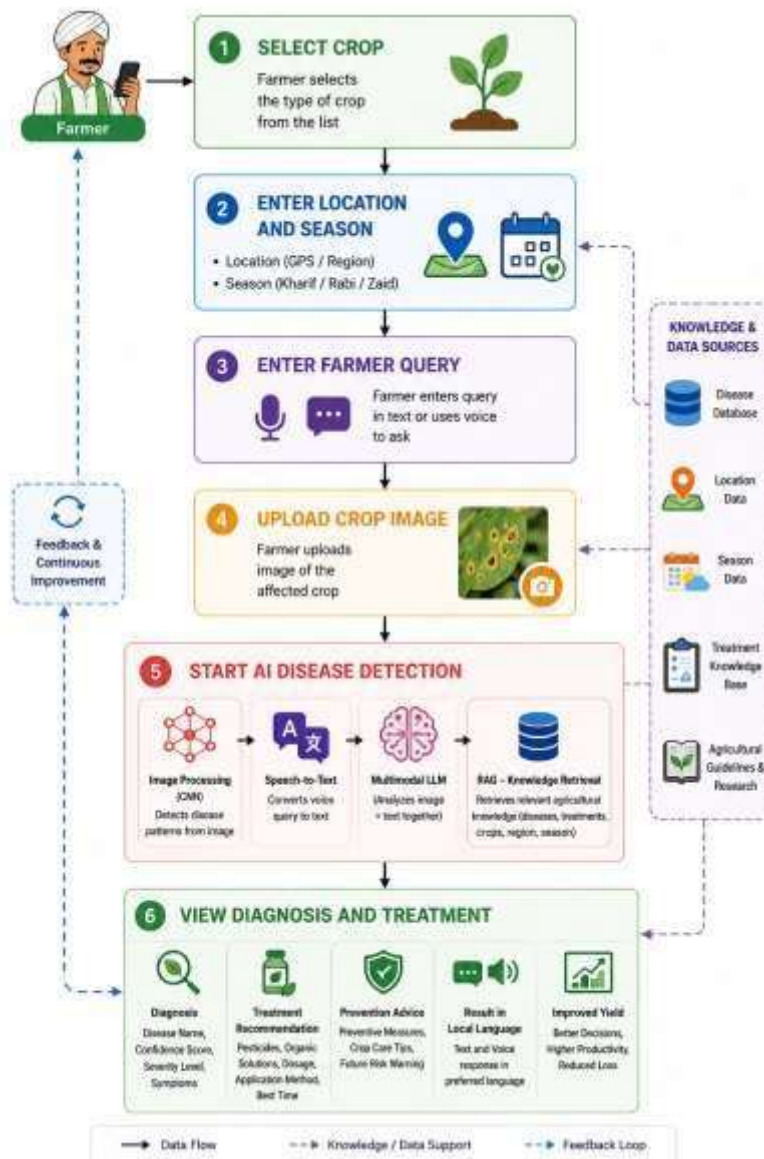


Figure 3.1: Proposed System Architecture (from project presentation)

CHAPTER 4: IMPLEMENTATION AND MODULE-WISE CONTRIBUTION

This chapter presents the implementation of the Generative AI System for Crop Disease Diagnosis and Localized Treatment Recommendation. It covers the software environment, system workflow, all functional modules, detailed implementation of Module 1 as the primary contribution of Mounish A, implementation evidence from the supplied presentation, testing and individual contributions. **4.1 Hardware and Software Environment**

Component	Specification / Purpose
Programming Language	Python
Application Framework	Streamlit
AI Model / Service	Multimodal LLM / Gemini as represented in the project presentation
Voice Processing	Speech-to-Text for farmer voice queries
Image Input	JPG, PNG and WEBP crop images
Development Environment	Visual Studio Code
Visualization	Streamlit interface and analytics charts

Table 4.1: Hardware and Software Environment

4.2 System Architecture and Implementation Workflow

The implemented workflow is organized as Crop Information → Image Upload → Farmer Query → Voice-to-Text when required → Multimodal AI Diagnosis → Recommendation → Analytics / Assistance. The architecture is modular so that the user-input layer can be validated independently from the AI reasoning layer.

- Module 1 collects and validates crop context and image input.
- Module 2 handles text/voice queries and multimodal diagnosis.
- Module 3 presents disease analytics and recommendation information.
- The implementation evidence supplied in the PPT demonstrates the Streamlit interface and functional screens.

4.3 Module 1 – Crop Information & Image Upload Module (Mounish A)

Module 1 Overview

Module 1 forms the primary user-input and context-preparation layer of the proposed Generative AI system. It is responsible for collecting the information required before multimodal reasoning begins. The module captures the crop type, crop location, current season and plant image in a structured interface. These inputs provide the AI system with visual evidence together with contextual information, improving the quality and relevance of subsequent diagnosis and recommendation.

Professional Design Rationale

The module is designed as a simple, guided workflow so that a farmer can provide all essential information without navigating complex technical controls. Crop type identifies the plant category, location establishes the geographical context, season adds environmental context, and image upload supplies the primary visual evidence. The presentation specifies support for JPG, PNG and WEBP image formats.

Input Validation and Data Integrity

Before the information is passed to the multimodal AI stage, the module checks that the required fields are available and that a supported image has been supplied. This prevents incomplete requests from entering the diagnosis workflow. Maintaining synchronization between the selected crop, location, season and uploaded image is an important design consideration because incorrect context can reduce the usefulness of the generated response.

Module 1 Integration

After successful input collection and validation, Module 1 hands the prepared context to the multimodal diagnosis stage. The complete information package can be represented as: Crop Type + Location + Season + Plant Image →

Validated Multimodal Input → AI Diagnosis and Recommendation. Thus, Module 1 is not only an interface component; it is the foundation for the downstream Generative AI workflow.

4.3.1 Module Objectives

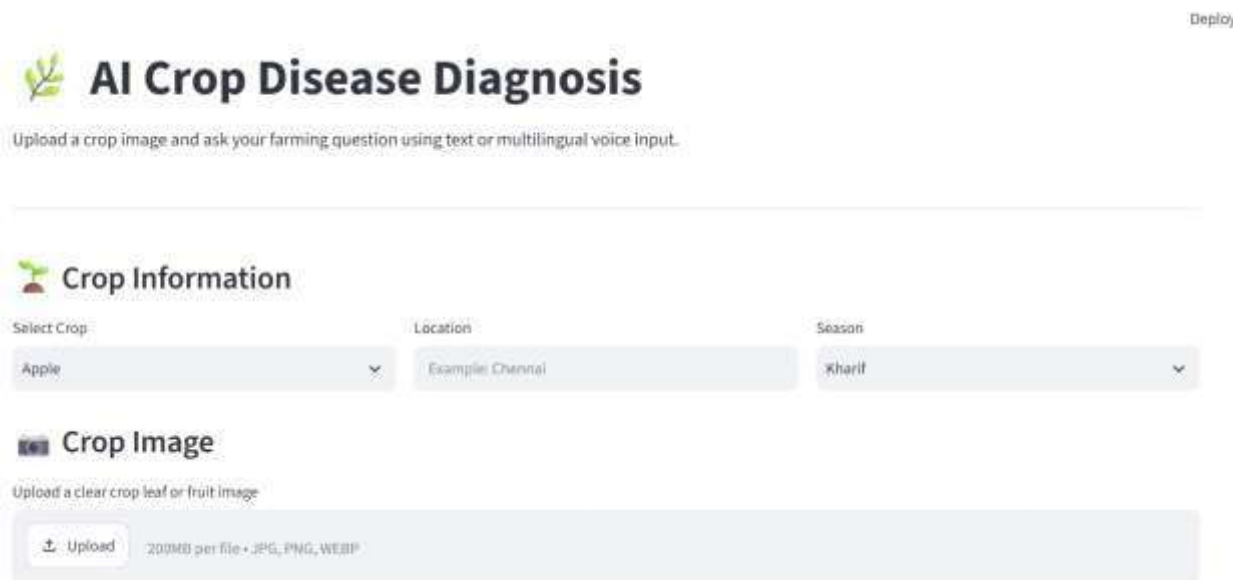
- Provide a clear crop-selection interface.
- Capture the location of the crop for localized recommendations.
- Capture the current season as additional context.
- Allow image upload in common formats such as JPG, PNG and WEBP.
- Validate required fields before diagnosis.
- Pass complete input context to the multimodal AI stage.

4.3.2 Functional Flow

- Open the crop disease diagnosis application.
- Select the crop type.
- Enter or select the crop location.
- Select the current season.
- Upload the crop image.
- Validate that required information is present.
- Use the prepared input as the context for multimodal analysis.

4.3.3 Design and Development

The Module 1 interface shown in the project presentation uses a structured form so that the farmer can provide crop information before uploading the image. The design reduces ambiguity by separating crop type, location, season and image selection. This is important because the multimodal model should receive consistent context rather than depending only on visual appearance.



The screenshot displays the 'AI Crop Disease Diagnosis' web interface. At the top, there is a title 'AI Crop Disease Diagnosis' with a small plant icon and a subtitle 'Upload a crop image and ask your farming question using text or multilingual voice input.' Below this, the 'Crop Information' section contains three dropdown menus: 'Select Crop' (set to 'Apple'), 'Location' (set to 'Example Chennai'), and 'Season' (set to 'Kharif'). The 'Crop Image' section below it has a subtitle 'Upload a clear crop leaf or fruit image' and an 'Upload' button with a file size limit of '200MB per file • JPG, PNG, WEBP'.

Figure 4.1: Module 1 – Crop Information and Image Upload Interface

4.3.3 Design and Development – Implementation Evidence

The supplied presentation provides implementation evidence for the crop-information screen. The interface includes a crop-selection field, location field, season field and an image-upload area. This screen represents the first operational stage of the system and is therefore given priority in this report. **Module 1 Data Preparation Logic**

Input Field	Role in AI Context	Validation / Expected Behaviour
Crop Type	Identifies the crop being analyzed	Required selection
Location	Supports localized recommendation	Required / meaningful entry
Season	Adds contextual information	Selection from available options
Crop Image	Primary visual evidence	Supported image format and valid upload

Table 4.2: Module 1 Input Responsibilities

4.3.4 Module 1 Responsibilities

Function	Implementation Work	Expected Behaviour
Crop Selection	Collect crop type from the user	Selected crop becomes AI context
Location	Collect crop location	Location is available for localized reasoning
Season	Collect current season	Season is available for contextual interpretation
Image Upload	Accept crop image	Valid image becomes multimodal input
Validation	Check required fields and supported image formats	Invalid/incomplete input receives feedback
Context Handoff	Prepare structured information for next module	Diagnosis stage receives complete context

Table 4.3: Module 1 Functional Responsibilities

4.3.5 Testing of Module 1

Test Case	Scenario	Expected Result	Status
TC-M1-01	Valid crop selection	Crop value accepted	Functional test
TC-M1-02	Location supplied	Location retained for context	Functional test
TC-M1-03	Season selected	Season retained for context	Functional test
TC-M1-04	Valid image uploaded	Image accepted for analysis	Demonstrated in PPT
TC-M1-05	Unsupported/invalid input	User receives validation feedback	Required behaviour
TC-M1-06	All inputs supplied	Complete context proceeds to diagnosis	Functional workflow

Table 4.4: Module 1 Functional Test Cases

4.4 Module 2 – Multimodal AI Diagnosis & Farmer Query Module

Module 2 allows farmers to ask questions through text or voice. The supplied PPT shows a voice-input interface with a language selection area, recording control and farmer-query field. Speech-to-Text converts the spoken query into text, after which the crop image, structured crop information and query are provided together to the multimodal AI service.

- Text input for farmer questions.
- Voice input for natural interaction.
- Speech-to-Text conversion.
- Preferred voice-language selection where implemented.
- Combination of image + crop information + farmer query.
- Multimodal AI/Gemini interpretation.



Figure 4.5: Farmer Voice / Query Interface Evidence

4.5 Module 3 – Disease Analytics & Recommendation Module

Module 3 presents the information generated from diagnosis in a form that can be understood and reviewed. The project presentation includes a Crop Disease Analytics screen showing total diagnoses, crop/disease counts, average confidence and a disease-distribution chart.

- Display total diagnoses performed.
- Show crop and disease counts.
- Present average model confidence where available.
- Visualize disease distribution using a chart.
- Connect diagnosis information to recommendation and preventive guidance.

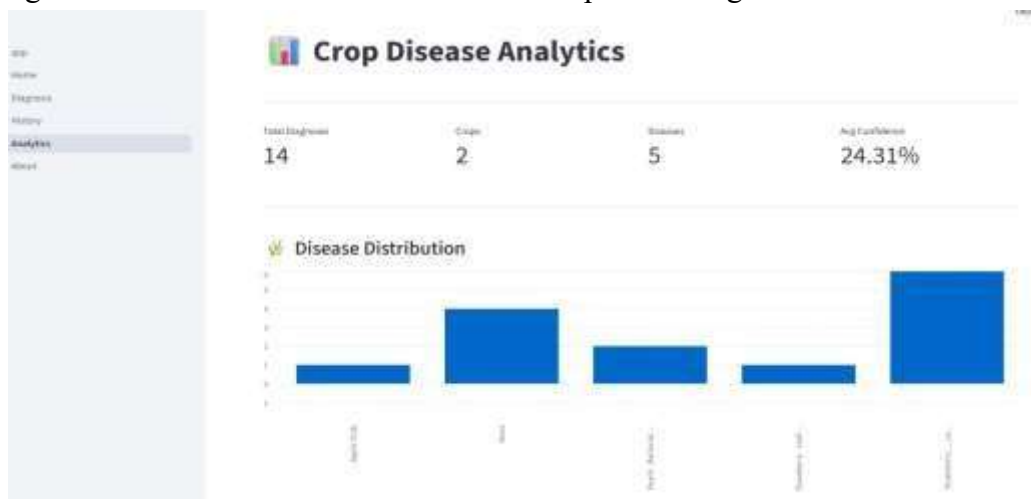


Figure 4.6: Crop Disease Analytics Dashboard Evidence

4.6 Implementation Evidence and Integration

The supplied presentation contains implementation evidence for the main user interaction screens. The evidence demonstrates that the system is not limited to a conceptual architecture: it includes an operational Streamlit interface for crop information, voice interaction and analytics.

4.7 Team Contribution and Module Ownership

The project was developed collaboratively. Mounish A is responsible for Module 1, Surya Krishna Teja contributes to the multimodal AI diagnosis and farmer query workflow, and N. Somasekhar Naidu contributes to analytics, recommendations and user-facing result support.

4.8 Individual Contribution – Mounish A

Mounish A's main contribution is the design and development of Module 1. The work includes the crop information interface, location and season fields, image-upload workflow, input validation and the preparation of complete context for downstream multimodal diagnosis. The implementation screenshots in Figures 4.1–4.7 provide visual evidence of this contribution.

CHAPTER 5: TESTING, RESULTS AND ANALYSIS

5.1 Testing Strategy

Testing is organized around Module 1 input functionality, voice interaction, multimodal processing, analytics presentation and the end-to-end workflow. The supplied project material demonstrates the interface and workflow, while exact numerical diagnostic accuracy should be reported only from a controlled evaluation dataset.

Test ID	Test Scenario	Expected Result	Status
TC01	Select crop and provide location	Input context accepted	PASS / functional
TC02	Select season	Season becomes context	PASS / functional
TC03	Upload supported crop image	Image accepted	PASS / demonstrated
TC04	Ask farmer question through text	Question passed to AI stage	PASS / functional
TC05	Provide voice query	Speech converted to text	Functional test
TC06	Combine image and query	Multimodal analysis requested	Functional test
TC07	View analytics	Statistics and chart displayed	Demonstrated
TC08	View recommendation	Localized guidance displayed	Functional test

Table 5.1: Functional Test Cases

5.2 Sample End-to-End Output

Representative flow: a farmer selects the crop, enters location and season, uploads a crop image, and asks a question using voice or text. The system converts voice to text when required, combines the inputs and sends them to the multimodal AI stage. The output is expected to describe the probable disease or symptoms and provide appropriate preventive or treatment guidance in understandable language.

5.3 Results

- The project demonstrates an integrated workflow for crop information, image input, farmer query and AI-based assistance.
- Module 1 provides a clear and structured entry point for the AI system.
- The multimodal design combines image evidence with crop and farmer context.
- Voice interaction supports natural communication with the system.
- The analytics interface provides a visual summary of diagnosis activity.
- The system generates localized and crop-specific recommendations as represented in the project presentation.

5.4 Visual Result Evidence 5.5 Result Interpretation

The supplied presentation supports the conclusion that the application includes crop information, image upload, multimodal diagnosis, voice interaction and analytics screens. However, the presentation does not provide a validated benchmark dataset or controlled numerical accuracy experiment. Therefore, this report does not claim an unsupported accuracy value.

5.4 Strengths

- Multimodal interaction combines visual and textual context.
- Module 1 provides structured inputs before AI reasoning.
- Voice input improves accessibility for users who prefer speaking.
- Generative responses can be expressed in simple, user-friendly language.
- Location and season create a foundation for localized recommendations.
- Streamlit enables rapid prototyping and demonstration.

5.5 Limitations

- Model responses can be affected by image quality and ambiguity of symptoms.
- Generative AI output may require expert verification for high-impact treatment decisions.
- Disease coverage depends on model knowledge and the available evaluation data.
- Speech recognition may be affected by background noise, pronunciation and language support.
- Real-time weather and soil context is not yet integrated.
- A larger, balanced crop-disease dataset is required for rigorous benchmarking.

5.6 Performance Evaluation Requirements

For a future controlled evaluation, the project should report disease-class accuracy, precision, recall and F1-score using a held-out test set. Additional evaluation should measure response latency, imageprocessing time, voice transcription accuracy and user satisfaction.

CHAPTER 6: CONCLUSION, FUTURE SCOPE AND REFLECTION

6.1 Conclusion

The Generative AI System for Crop Disease Diagnosis and Localized Treatment Recommendation demonstrates how multimodal large language models can be combined with crop images, structured information and farmer voice/text queries to create an accessible agricultural assistance platform.

The project integrates crop information capture, image upload, Speech-to-Text interaction, multimodal AI analysis, disease interpretation, localized recommendations and analytics. Module 1 is particularly important because reliable AI assistance depends on receiving complete and meaningful crop context before analysis.

6.2 Future Scope

- Improve disease diagnosis through larger and more diverse crop-image datasets.
- Add stronger support for Telugu, Tamil, Hindi and other regional languages.
- Integrate real-time weather, soil and agricultural advisory data.
- Provide location-based and season-specific recommendation rules.
- Add continuous voice conversations with farmers.
- Add confidence thresholds and expert-review options for uncertain cases.
- Develop a mobile application for field-level access.
- Introduce explainable AI and evidence-backed recommendations.

6.3 Student Reflection

The project provided practical experience in Generative AI, multimodal interaction, Streamlit application development, image-based input handling, voice processing, interface design, testing and technical documentation. The detailed work on Module 1 demonstrated that AI performance is influenced not only by the model but also by the quality, completeness and usability of the information supplied to it.

PROJECT IMPLEMENTATION EVIDENCE

The following figures are taken from the supplied project presentation and document the implementation of the proposed Generative AI application.

Voice Input

Select Voice Language

English

Select the language you are speaking, then record your complete farming question.

Record your farming question

00:00

Farmer Query

Ask your question

Example: What disease is affecting my apple plant and how can I treat it?

START DISEASE DETECTION

Deploy

AI Crop Disease Diagnosis

Upload a crop image and ask your farming question using text or multilingual voice input.

Crop Information

Select Crop	Location	Season
Apple	Example: Chennai	Kharif

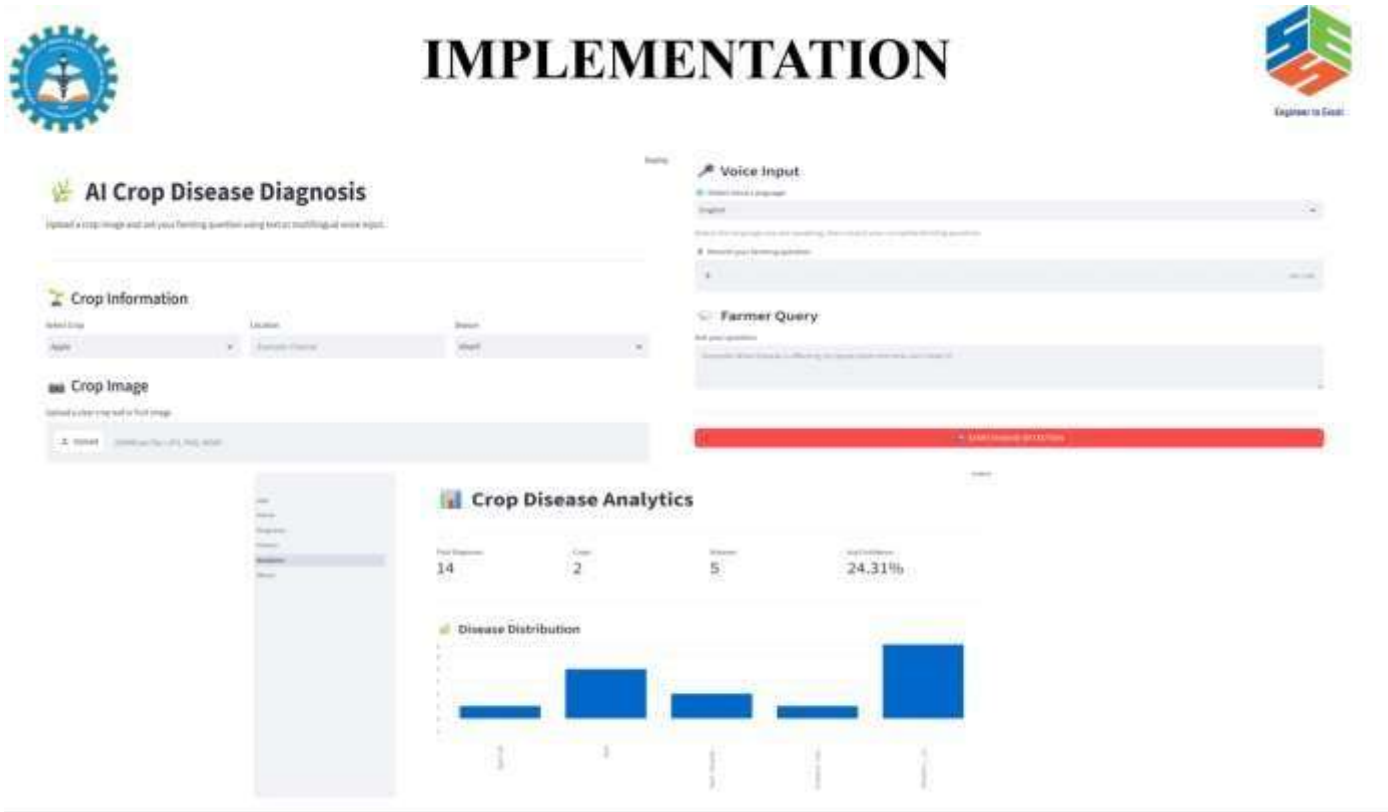
Crop Image

Upload a clear crop leaf or fruit image

Upload 200MB per file • JPG, PNG, WEBP

PROJECT IMPLEMENTATION EVIDENCE – ADDITIONAL SCREENS

These implementation visuals support the functional descriptions presented in Chapter 4. The architecture diagram in Figure 3.1 and the Module 1 screens in Figures 4.1–4.2 provide the main evidence for the primary contribution.



REFERENCES

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APPENDIX A: REPRESENTATIVE IMPLEMENTATION NOTES

A.1 Module 1 Input Handling

The implementation logic for Module 1 can be represented as the sequence: select crop → enter location → select season → upload image → validate → prepare context → invoke diagnosis stage. The supplied PPT demonstrates the corresponding interface.

A.2 Multimodal Request Structure

A multimodal request logically combines the uploaded image with structured crop information and the farmer's query. The AI service is expected to use all available context rather than relying on image appearance alone.

A.3 Voice Interaction

The voice workflow captures farmer speech, converts it to text and passes the resulting query to the multimodal processing stage. The presentation includes a dedicated voice-input screen demonstrating this interaction.

A.4 Recommendation Output

- Probable disease or symptom interpretation.
- Important visual or contextual indicators.
- Localized treatment considerations.
- Preventive actions.
- Advice to seek agricultural expertise when uncertainty is high.

A.5 Implementation Notes

- Validate image type and availability before analysis.
- Keep crop context synchronized with the uploaded image.
- Handle speech-recognition failures gracefully.
- Do not expose API credentials in source code.
- Log only information necessary for application debugging and evaluation.

APPENDIX B: FIGURE AND SCREENSHOT INDEX

Figure	Description
Figure 3.1	Proposed system architecture showing crop information, multimodal diagnosis, analytics/recommendation and user interaction flow.
Figure 4.1	Module 1 – Crop Information and Image Upload interface.
Figure 4.2	Module 1 – Implemented input screen from the project presentation.
Figure 4.5	Farmer voice / query interface evidence.
Figure 4.6	Crop Disease Analytics dashboard evidence.
Figure 5.1	Crop Disease Analytics result screen.
Figure 5.2	Farmer voice / field assistance visual.
Figure 7.1	Crop Information and Image Upload implementation evidence.
Figure 7.2	Farmer Voice Query implementation evidence.
Figure 7.3	Crop Disease Analytics dashboard.
Figure 7.4	Farmer Voice / Field Assistance visual.

Table B.1: Figure and Screenshot Index